SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Total Marks:50**

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I V. Pradhyumna (192425336) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

95/100

Annexure A

APPLICATION- BASED QUESTIONS

1. Scenario: Event Communication Security: A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message "MEET AT NOON" is transformed into an encoded version using a fixed shift pattern.

a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
b. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

2. Scenario: Intercepted Suspicious Message: A cybersecurity intern intercepts the coded message: "GSRH RH Z HVXIVG" during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

a. Deduce the most likely original message.
b. Justify your answer by analysing repeating patterns, letter structures, or known cipher characteristics.

3. Scenario: Secure Messaging System: A company deploys a keyword-based encryption technique for internal communication. An employee sends the word "HELLO" using the keyword "KEY" to ensure confidentiality.

a. Demonstrate how the encryption process transforms the message step-by-step.
b. Explain how the repeating keyword influences each character of the ciphertext.

4. Scenario: Suspicious Digital Asset Investigation: During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.

a. Analyze how investigators could detect the presence of hidden data within the image.
b. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

5. Scenario: Legacy Encryption in Corporate Communication: A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: "WECRLTEERDSOEEFEAOCAIVDEN".

a. Determine the most probable original message by reversing the encryption process.
b. Critically evaluate why such a method is vulnerable in modern communication systems.
c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

ENGINEERING

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

2 Intercepted Suspicious Message

Scenario

A cyber security intern intercepts the coded message:

GSRH RH ZHVXIVG

The structure of the message suggests that a simple substitution cipher has been used rather than a complex encryption algorithm.

a) Deduce the most likely original message

The given ciphertext is:

GSRH ZHVXIVG

The most likely encryption technique is the Atbash cipher

Athash substitution Table

Plaintext	A	B	C	D	E	G	H	I	J	K	L	M	N	O
ciphertext	N	O	P	Q	S	T	U	V	W	X	Y	Z	M	L

Plaintext	P	Q	R	S	T	U	V	W	X	Y	Z
ciphertext	K	J	I	H	G	F	E	D	C	B	A

word 1: GSRH

Using Re Atbash mapping:

G → T

S → H

R → I

H → S

word 2: RH

R → I

H → S

word 3: Z

Z → A

word 4: HVXIVG

H → S

V → E

X → C

I → R

V → E

G → T

Final decrypted message

GSRH RH Z HVXIVG

THIS IS A SECRET

6) Justification of the answer

The answer can be justified by examining the letter patterns, word lengths, and substitution characteristics.

1. Word-length analysis

The ciphertext consists of four words:

Ciphertext	No. of letters
GISRH	4
RH	2
Z	1
HIUXIVGI	6

2. One-to-one substitution

Atbash uses a fixed substitution relationship

for exc

G ↔ T
S ↔ H
R ↔ I
H ↔ S

The same mapping is consistently applied throughout the message.

3. Recognizable word structure:

The ciphertext:

GSRH RH Z HVXIVG

has a natural sentence structure:

4letters + 2letters + 1 letter + 6letters

The decrypted result:

THIS IS A SECRET

forms a grammatically meaningful English sentence.

4. Repeating letters

The word HVXIVG contains the pattern:

H V X I V G

the letter V occurs twice in positions 2 and 5.

After Atbash substitution:

S E C R E T

Conclusion

The intercepted message is most likely encrypted using the Atbash substitution cipher. Since Atbash reverses the alphabet, applying the same substitution again successfully recovers the plaintext:

THIS IS A SECRET

Digital Asset Investigation

Scenario

During a digital forensic audit, investigators suspect that a company logo image contains hidden confidential information. The image appears completely normal during casual inspection.

(a) How investigators can detect hidden data

Investigators can perform steganalysis, which is the process of examining digital media for signs of hidden information.

A systematic investigation can be carried out as follows:

Step 1: Examine the original image

The investigator first records basic information about image:

- File Format
- File size
- Image dimensions
- Colour depth
- creation / modification information
- Meta data

However, file size alone does not prove that hidden information exists.

Step2: visual inspection

The image can be enlarged and carefully inspected.

Investigators may look for:

- unusual pixel patterns
- unexpected colour changes
- strange blocks or regions
- visible distortions
- irregular textures
- differences between apparently similar areas

Steganography is normaly designed to make changes difficult to notice.

Step 3: Histogram analysis

A histogram shows the distribution of pixel intensity or colour values in a image.

Investigators can compare the histogram of the suspicious image with:

- the expected distribution for similar inges
- A known original version of the logo
- another copy of the same image

Unusual patterns in the histogram can indicate that pixel values have been manipulated.

step 4: Analyze least-significant Bits (LSB)

One common image-steganography approach is to hide information in the least significant bits of pixels.

A pixel may contain values such as:

Red = 10110110

Green = 01101001

Blue = 11001010

The least significant bit is the rightmost bit:

Red = 1011011[0]

Green = 0110100[1]

Blue = 1100101[0]

step 5: Examine metadata

Investigators can inspect metadata associated with the image.

Examples include:

- file creation information
- modification information
- software used to create/edit the image
- comments
- embedded metadata fields.

Suspicious metadata does not necessarily prove steganography but it can provide useful forensic evidence.

(b) Two techniques to strengthen hidden communication

(1) Encrypt the data before embedding

confidential message
↓
AES Encryption
↓
Ciphertext
↓
Embed inside image
↓
Final image

(2) use randamized / adaptive embedding

Secret data
↓
Secret key
↓
Select suitable / randomized pixels
↓
Embed information
↓
stego image

Final conclusion:

The suspicious logo can be investigated using visual inspection, metadata examination, histogram analysis, LSB analysis, comparison with the original image and steganalysis tools.